

Preventing disease emergence following eradication: Application to mpox

Supplemental Information

1 Multi-type branching process formulation

Our model for the adaptive evolution of a zoonotic pathogen after spillover into humans is based on a previously developed multi-type branching process model of emerging infectious diseases [1]. The multi-type branching process formulation allows for an easy derivation for the probability of emergence.

Consider the case where a pathogen must accumulate a number of mutations after spillover into the human population to attain $R_0 > 1$. Given m variants from the wildtype to the fully-evolved mutant with $m - 1$ mutations required to attain $R_0 > 1$, the m -type branching process can be described by m probability generating functions (PGFs):

$$f_i(s_1, s_2, \dots, s_m) = \sum_{j_1, j_2, \dots, j_m=0}^{\infty} p_i(j_1, j_2, \dots, j_m) s_1^{j_1} s_2^{j_2} \dots s_m^{j_m}, \quad i \in \{1, 2, \dots, m\}$$

where $p_i(j_1, j_2, \dots, j_n, \dots, j_m)$ is the probability that an infection of type i gives rise to j_n secondary infections of type n .

The probability of extinction q_i of a transmission chain initiated by a single infection with a variant of type i is given by the least fixed-point solution to: $f_i(q_1, q_2, \dots, q_m) = q_i$, $i \in \{1, 2, \dots, m\}$.¹

We define a model in which there exists m different variants of the pathogen, with one wildtype (spillover), one fully evolved, and $m - 2$ intermediate mutants. We denote the basic reproductive number of the wildtype pathogen ($R_{0,Spillover}$) as $R_0^{(1)}$, the basic reproductive numbers of the $m - 2$ intermediate mutants as $R_0^{(2)}, \dots, R_0^{(m-1)}$, and the basic reproductive number of the fully-evolved mutant ($R_{0,Evolved}$) as $R_0^{(m)}$.

We assume the following about mutation and the spread of infections:

¹Note that this is a special case of the generalized expression for the probability that a process initiated with $r_1, r_2, \dots, r_n$ chains of infections of type $1, 2, \dots, m$ goes extinct, which is given by the product: $q_1^{r_1} q_2^{r_2} \dots q_m^{r_m}$.

- the mutation rate, μ , of all the variants is the same
- only single variants occur, i.e the i th variant can only mutate into an $i + 1$ mutant
- the total number of secondary infections arising from an individual infected with variant i is Poisson-distributed with mean $R_0^{(i)}$
- the number of secondary infections of type i arising from an individual infected with variant i is Poisson-distributed with mean $(1 - \mu)R_0^{(i)}$
- the number of secondary infections of type $i + 1$ arising from an individual infected with variant i is Poisson-distributed with mean $\mu R_0^{(i)}$

These assumptions define the PGFs $f_1, f_2, \dots, f_m$, given by:

$$f_i(s_1, s_2, \dots, s_m) = \exp[(1 - \mu)R_0^i(s_i - 1)] \exp[\mu R_0^i(s_{i+1} - 1)], \quad i \in \{1, 2, \dots, m-1\}$$

$$f_m(s_1, s_2, \dots, s_m) = \exp[R_0^m(s_m - 1)]$$

The probability of emergence can be calculated numerically from the PGFs according to the fixed point equation above.

Our model makes two important departures from the previous model:

1. The fitness of intermediate variants increases by some proportional amount, σ ;
2. Evolution continues to increase the reproductive number after the first $R > 1$ variant.

We alter the framework presented in [1] according to the following.

Consider the case of adaptive evolution during transmission. After the spillover infection, mutations are acquired during subsequent transmission events, each increasing the R of infections by some proportion, σ , until a variant emerges with an $R_0^{(m)}$ exceeding some $R_0^{(m')}$.

The number of mutations $m - 1$ in the lineage from the wild-type (1) to the fully-adapted type (m) thus depends on the reproductive number of the wildtype $R_0^{(1)}$, the minimum reproductive number of the fully-adapted type $R_0^{(m')}$, and the proportional increase in $R_0^{(i)}$ to $R_0^{(i+1)}$ with mutation and is given by:

$$m - 1 = \frac{\log(R_0^{(m')}) - \log(R_0^{(m')})}{\log(1 + \sigma)}$$

Note that $R_0^{(m)}$ is the first R_0 that is greater than the minimum $R_0^{(m')}$.

To calculate the probability that the lineage does not go extinct before exceeding $R_0^{(m')}$, we replace the final $R_0^{(m)}$ with some R arbitrarily high such that the contribution of extinction of the final type is negligibly small (i.e once the final type emerges, the process will never go extinct, and so the calculated extinction probability is determined by the intermediates.)

Below, we illustrate the relationship between the additive fitness model in [1] and the above model in which the number of intermediate mutants decreases as R_0 increases. For the sake of comparison with the earlier model, we calculate the non-extinction probabilities for the branching processes before supercriticality ($R_0^{(m')} = 1$).

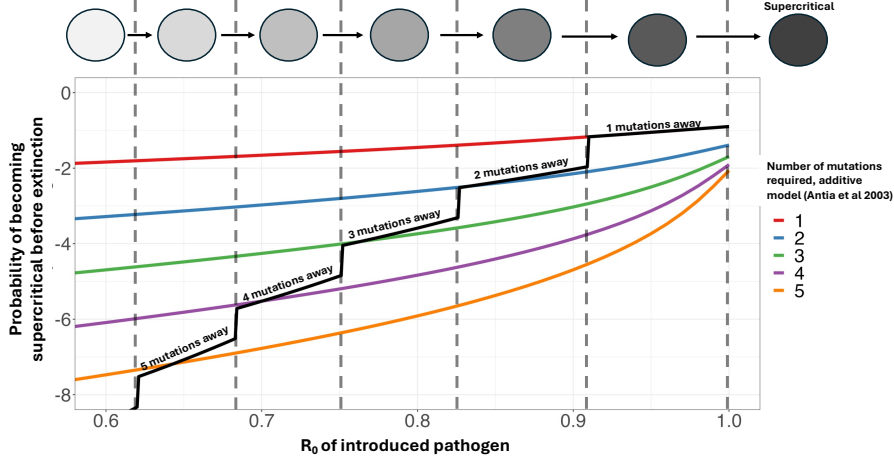


Figure 1: **Schematic of model adapted from [1].** Colored lines show additive model from [1], where a given number of mutations are required for the emergence of a variant with $R_0 > 1$, and R_0 increases with each mutation. The black line shows the current model, formulated such that the increase with each mutation is fixed and thus the number of mutations required for emergence decreases with increasing R_0 . The circles correspond to variants requiring a decreasing number of mutations for emergence.

The colored lines show the additive fitness model given in [1], given various numbers of mutations required for emergence. The earlier Antia model gives the probability of becoming supercritical given m mutations are required for emergence, regardless of the spillover R_0 - the increase in R with mutation decreases as R approaches 1.

Instead, the current model considers the increase with mutation to be a fixed proportion, σ , and so the number of mutations required for emergence decreases as $R_{0,Spillover}$ increases. As a result, P_E increases sharply where the spillover R has increased such that one fewer mutation is required for emergence (SI Fig 1).

2 Strong dependence of P_P on $R_{0,Spillover}$ is robust to μ and σ

We find that P_p changes much more sharply with linear increases in R_0 than s , due to the steep dependence of P_E on R_0 . This result is robust to a range of beneficial mutation rates, μ and increases in fitness with mutation, σ (SI Fig 2).

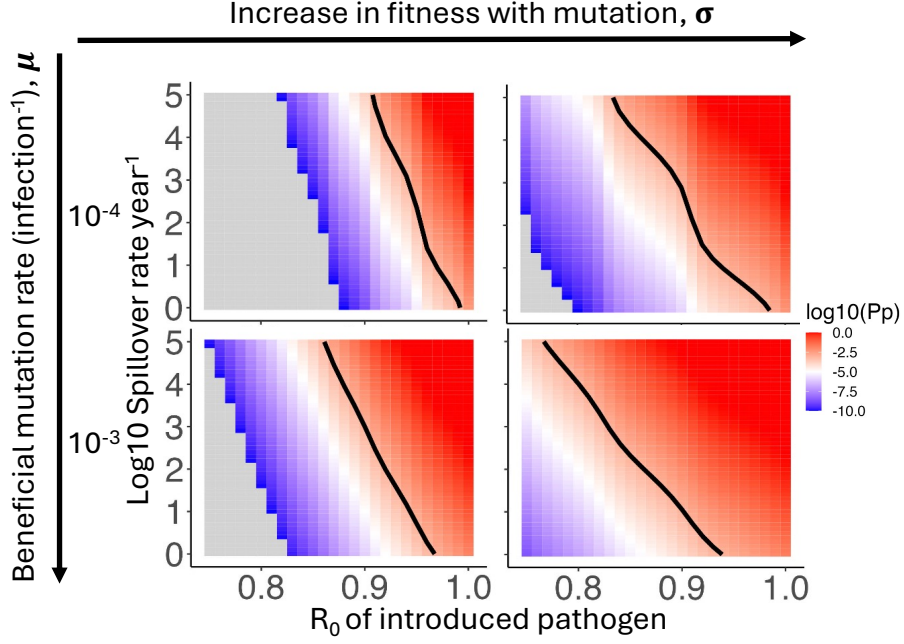


Figure 2: **Effect of σ and μ on the dependence of pandemic potential on R_0 and s .** The strong dependence of P_p on R_0 rather than s is robust to changes in the pathogen characteristics μ and σ .

3 Strong dependence of P_p on R_0 is also characteristic of a pre-adaptation pathway.

We now consider pre-adapted variants of a pathogen that circulate in the zoonotic reservoir. A pre-adapted pathogen variant might, for example, require one mutation for successful adaptation to transmission between humans, reflected by a large σ . The rate of spillover of this variant into humans would be proportional to their frequency in the zoonotic reservoir, and this is likely to be low (due to mutation-selection-drift balance). This is reflected in our model by log-fold decreases in s . Changes in pandemic potential are still largely driven by increase in R_0 rather than s along this pathway, with only a slightly reduced effect owed to the decrease in the number of mutations required for emergence (SI Fig 3).

4 Effect of epistasis and fitness valley on emergence probability.

We focus our exploration on the fitness landscapes that are most likely to give rise to highly transmissible variants - transmission increasing by a fixed propor-

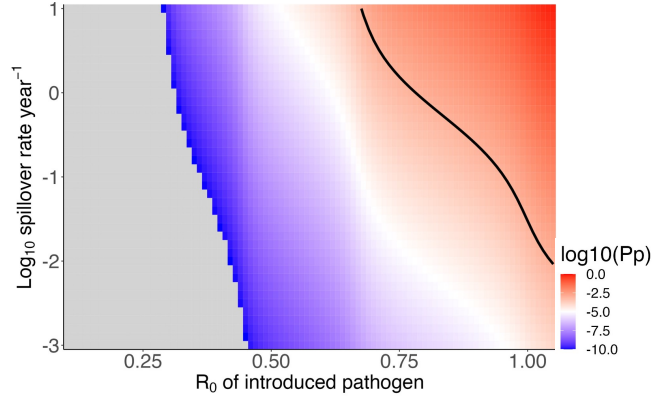


Figure 3: **In a pre-adapted pathway, variants with high R_0 and that are few infections away from emergence (due to large σ) may circulate endemically in the animal reservoir.** Assuming that the mutations conferring pre-adaptation to humans are not so deleterious that the variants cannot persist in the animal reservoir, they may circulate at low frequency (set by mutation-selection-drift balance.) The frequency of spillovers, s , would be proportional to the frequency in the zoonotic reservoir, and thus s would be in a significantly smaller regime. The dependence of pandemic potential on R_0 rather than s is maintained in this regime, and thus interventions targeting transmission may have the largest effect on preventing emergence, particularly when the magnitude of spillovers may be high and challenging to reduce on a log-scale. $c = 0.8$, $\mu = 10^{-3}$, $\sigma = 0.5$, and $s_0 = 10^{-3}$. Trajectories over time parameterized with demographic data from the DRC.

tion with each mutation, and low-frequency pre-adapted variants in the animal reservoir. Introducing epistatic interactions - where multiple mutations are required to achieve a fitness advantage, and some intermediate mutations may even introduce a deleterious fitness valley - is likely to dramatically reduce the probability that a subcritical variant becomes supercritical, to such an extent that, if a pandemic variant is to emerge in the human population and a consistently increasing or pre-adapted pathway exists, it is highly likely that the pathway to that emergence did not include epistatic interactions or fitness valley crossings.

References

- [1] R. Antia, R. R. Regoes, J. C. Koella, and C. T. Bergstrom. The role of evolution in the emergence of infectious diseases. *Nature*, 426(6967):658–61, 2003.